

Right now, data centers are growing at unprecedented rates.

Programs 26 [No author; Programs.com provide dozens of resources designed to help you find a degree program, pass a tough certification exam or get an interview for your first AI or cybersecurity role; “Measuring the Data Center Boom: Facts and Statistics (2026)”]; July 15, 2026
<https://programs.com/resources/data-center-statistics/> DOA: 8/14/2026]

Rapid adoption of AI among enterprises and consumers is fueling the growth of data centers, the massive structures that provide the necessary computing power.

The data center boom impacts nearly everyone, from the tech and real estate sectors to policymakers and the environmentally conscious, who respectively view these data centers as a geopolitical asset and a drain on natural resources.

This report covers the most critical statistics and trends regarding data centers in 2026.

Editor's Picks

- As of July 2026, there are 11,959 data centers located across 179 countries.
- Demand for data centers is predicted to nearly triple by 2030.
- Between now and 2030, companies worldwide are expected to invest nearly \$7 trillion in building and upgrading data centers.
- Global data center power usage is expected to increase to 219 GW over the next five years, enough to power roughly 180 million US homes.
- On average, data centers cover 100,000 square feet, but hyperscale data centers are as large as 10 million square feet.
- Approximately 38% of data centers are located in the US.
- Large data centers use up to 5 million gallons of water per day for cooling purposes.
- US states with multiple data centers create more than \$30 billion in additional economic output annually.

Current State & Future Growth of Data Centers

The scale, capacity, and market value of data centers are growing exponentially.

Much of this is due to the increasing demand for AI among enterprises and consumers. But countless other business operations (like processing data from IoT devices) also depend on data centers.

Increasing consumer demand is just as powerful. Streaming services, ecommerce apps, and social media are creating massive workloads for data centers to process.

There are approximately 12,000 live data centers around the world (Data Center Map)

As of July 2026, there are 11,959 active data centers worldwide.

The majority of these data centers are hyperscale.

Holland 26 Makenzie Holland [Makenzie Holland graduated from Indiana University Bloomington in 2015 with a Bachelor of Arts in Journalism. Prior to joining CIO Dive, she was a federal technology policy reporter at Informa TechTarget, a general assignment reporter for the Wilmington StarNews and a crime and education reporter at the Wabash Plain Dealer], April 8 2026, “Hyperscalers will own two-thirds of data center capacity by 2031”, ciodive

<https://www.ciodive.com/news/hyperscalers-two-thirds-data-center-capacity-2031/817016/>, DOA:

08-30-2026 //SC //recut RT september 18

- **Hyperscalers including Google, Microsoft and AWS will account for 67% of data center capacity by 2031, driven by significant AI infrastructure investments**, according to a report from Synergy Research Group published Tuesday. Enterprise on-prem data centers will drop from 56% of total data center capacity in 2018 to 19% by 2031, the firm said.
- “Almost 60% of that hyperscale capacity is now in own-built, owned data centers with the balance being in leased facilities,” according to the report. “With non-hyperscale colocation capacity accounting for another 20% of total capacity, that leaves on-premise data centers with just 32% of the total.”
- On-prem data center’s share of total capacity remained relatively constant over the last eight years, even seeing a boost from enterprise AI deployments and GPU servers, according to the report. **However, by 2031, hyperscalers will have 14 times the amount of data center capacity than 2018 as AI accelerates expansion.**

Dive Insight:

As enterprises rush to adopt AI, hyperscalers are building data centers to keep up with demand and navigate capacity crunches while facing rising concerns about electricity costs and community pushback.

The top three hyperscalers, Microsoft, Google and AWS, plan to invest more than \$500 billion in capital expenditures for infrastructure supporting AI deployment in fiscal year 2026. Hyperscalers operated 1,297 large data centers globally in Q3 2025, nearly triple the amount from 2018, according to a separate SRG report. That number rose to 1,360 by the end of 2025.

“We will increase our total AI capacity by over 80% this year and roughly double our total data center footprint over the next two years, reflecting the demand signals we see,” Microsoft CEO Satya Nadella said during the company’s Q1 2026 earnings call in October.

It’s not just hyperscalers vying for compute power. Large language model provider Anthropic earlier this week announced an expanded partnership with Google to secure its TPU chips, supplied by Broadcom, adding multiple gigawatts of compute capacity starting in 2027. Meanwhile, LLM provider OpenAI in October partnered with AMD to add six gigawatts of AI compute capacity.

Despite massive investments to expand compute capacity, the tech industry is already facing shortages. During Microsoft’s Q1 2026 earnings call, EVP and CFO Amy Hood said Microsoft was struggling with a capacity shortage expected to extend throughout the fiscal year.

Naveen Chhabra, principal analyst at Forrester, previously told CIO Dive in an email that CIOs will need to prepare for compute capacity being a constrained resource.

In addition to capacity shortages, hyperscalers are grappling with concerns about how data centers will raise electricity costs. The U.S. Energy Information Administration said data center electricity demand could drive price hikes of up to 79% in areas like Texas by 2027.

The concern has not gone unnoticed. OpenAI in January committed to paying its own way on energy across its Stargate projects, a \$500 billion AI infrastructure push announced by President Donald Trump in January 2025.

Last month, the White House announced the Ratepayer Protection Pledge, a voluntary commitment from tech providers including Google, Oracle, xAI, Meta, Microsoft, OpenAI and Amazon to help offset electricity costs as data center buildouts continue.

The tech industry is also facing pushback on data center builds from local and state governments. A city in Wisconsin recently passed an anti-data center referendum, according to Politico reporting. Meanwhile, the Wall Street Journal reported that Maine could freeze data center construction until November 2027.

However, these data centers have detrimental impacts on society— from rising costs to water consumption. That’s why this expansion is opposed by a large majority of Americans, with By Jeffrey M. Jones [Jeffrey M. Jones, Ph.D., has served as a Gallup Senior Editor since 2000, overseeing research and conducting analysis for Gallup's U.S. polling and other public release surveys. His research on public opinion and voting behavior has been published in academic journals and edited books.], May 13th, 2026, “Americans Oppose AI Data Centers in Their Area”, Gallup, <https://news.gallup.com/poll/709772/americans-oppose-data-centers-area.aspx>, DOA: 08/09/26 //RT

WASHINGTON, D.C. — **Seven in 10 Americans oppose constructing data centers** for artificial intelligence in their local area, including nearly half, **48%, who are strongly opposed.** Barely a quarter favor these projects, **with 7% strongly in favor.**

These results, from a March 2-18 Gallup survey, represent the first time Gallup has asked about data center construction, a topic that has met **fierce opposition from local residents in many parts of the country.** These data centers house computing equipment that helps power AI technology used by businesses, universities and other institutions. The centers cover large areas of land, **require extensive amounts of electricity to operate and need substantial water** to cool the equipment, raising concerns about their impact on the environment and local electric bills.

The data center question parallels the wording Gallup uses to ask about local nuclear power plant construction. In the same March survey, 53% of Americans say they oppose building a nuclear energy plant in their area, far less than the 71% opposed to data center construction. Since Gallup first asked the nuclear power plant question in 2001, the high point in opposition has been 63%.

Opposition Tied to Environmental, Quality-of-Life Concerns

The March survey asked people to rate their level of concern about the environmental impact of AI data centers. Forty-six percent say they worry a great deal and 24% a fair amount, largely mirroring the degrees of opposition to data center construction.

To gain a deeper understanding of people’s reasons for favoring or opposing data center construction in their area, Gallup asked an open-ended question on an April web survey using the Gallup Panel. Americans who favor the building of a data center in their area mostly cite the potential economic benefits. Opponents of data centers have more varied reasons for their position, but they focus mostly on environmental concerns.

Half of opponents mention data centers’ excessive use of resources, including 18% each mentioning their use of water and energy. Sixteen percent mention a related environmental concern of pollution, including noise pollution and air and water pollution.

About one in five opponents are concerned with the impact on local quality of life, including increased population, increased traffic and preferring that the land be used for other purposes. A similar share mention potentially negative economic consequences, including higher utility bills, cost-of-living increases, and the cost of building the data centers (which could involve the use of taxpayer funds).

Most of the remaining opposition stems from general or specific concerns about artificial intelligence.

Meanwhile, two-thirds of those in favor of building data centers in their area cite the economic benefits, including 55% who mention increased job opportunities specifically. Others mention increased tax revenue (13%), housing and infrastructure development, and general economic benefits.

Smaller shares of data center proponents mention the potential benefits of AI to technology or themselves, personally.

Majority of Democrats Strongly Oppose Data Centers

Majorities of all major demographic groups, including all party groups, say they would oppose having a data center built where they live.

However, Democrats are much more likely than Republicans to be strongly opposed, 56% vs. 39%, with independents between the two at 48%.

Women (55%) are also more likely than men (43%) to register strong opposition to data center construction.

There are no meaningful differences in total opposition by age, race, education, income or urbanicity. Opposition is slightly lower among those living in the West (63%) and East (68%) than in the Midwest (76%) and South (75%).

Environmental concerns are a bigger source of differences in opposition to data center construction. Seventy-eight percent of U.S. adults who worry about the quality of the environment oppose data center construction, compared with 52% who are not worried.

That is a far bigger gap in opposition to data centers than is seen among those who are worried (73%) versus not worried (65%) about the availability and affordability of energy.

Thus Rebecca and I affirm resolved: The United States federal government should enact a moratorium on hyperscale data center construction.

Our first argument is electricity prices.

Hyperscalers create unprecedented demand.

From the **Department of Energy**, [department in the US federal government, highly reputable], no author, December 20th 2024, "DOE Releases New Report Evaluating Increase in Electricity Demand from Data Centers", Energy.gov,

<https://www.energy.gov/articles/doe-releases-new-report-evaluating-increase-electricity-demand-data-centers> DOA: 09/18/26 //RT

data centers could consume as much as 12% of electricity by 2028.

WASHINGTON, D.C. — The U.S. Department of Energy (DOE) today announced the publication of the 2024 Report on U.S. Data Center Energy Use produced by Lawrence Berkeley National Laboratory (LBNL) which outlines the energy use of data centers from 2014 to 2028. The report estimates that data center load growth has tripled over the past decade and is projected to double or triple by 2028. U.S. electricity demand is projected to account for data center expansion and the rise of artificial intelligence (AI) applications, domestic manufacturing growth, and electrification of different industries. DOE has anticipated this growing demand trend — it reflects robust industrial investments in America and national leadership on technology innovation. The Department continues to develop advanced technologies and leverage its resources to meet rising electricity demand in the United States while maintaining a reliable, affordable, and secure national energy system.

"The United States has seen an incredible investment in artificial intelligence and other breakthrough technologies over the last decade and a half, and this industrial renaissance has created greater demand on our domestic energy supply," said U.S. Energy Secretary Jennifer M. Granholm. "We can meet this growth with clean energy. Lawrence Berkeley National Laboratory's report on data center energy usage crucially underscores why the Department of Energy has developed and is deploying technologies to enable continued economic growth across American industries."

The report finds that **data centers** consumed about 4.4% of total U.S. electricity in 2023 and **are expected to consume** approximately 6.7 to **12% of total U.S. electricity by 2028.** The report indicates that total data center electricity usage climbed from 58 TWh in 2014 to 176 TWh in 2023 and estimates an increase between 325 to 580 TWh by 2028.

DOE is leveraging its resources to meet increasing electricity demand while improving critical infrastructure and advancing American economic competitiveness. DOE resources span the entire power system, from new generation and storage technologies to enhancing and expanding the transmission system to maximizing efficiency and flexibility of demand resources. DOE's key strategies for meeting data center energy demand include:

Enabling data center flexibility through onsite power generation and storage solutions, including the Industrial Efficiency and Decarbonization Office's Onsite Energy Program and new Industrial Energy Storage Systems Prize, so data centers can be a grid asset rather than a burden.

Leveraging energy community opportunities to re-use infrastructure at retired coal facilities for data centers and associated power infrastructure.

Engaging with stakeholders on innovative rate structures to support data center expansion while maintaining affordability.

Commercializing key enabling technologies such as next-generation geothermal, advanced nuclear, long-duration storage, and efficient semiconductor technologies.

Consumers are the ones paying to serve that demand.

Look to Virginia and Texas, home to more than 40% of the world's hyperscale data centers.

Stand Earth 26 Stand.earth [We are a growing team of strategists, researchers, communicators, policy and issue experts — plus a powerful global network of donors, activists, and community members from every walk of life, united in our demand for transformational change], no author, "Memo: Virginia residents sour on data centers, according to several recent polls", June 9 2026, <https://stand.earth/insights/memo-virginia-residents-sour-on-data-centers-according-to-several-recent-polls/>, DOA: 08-24-2026 //SC //recut RT

Virginia, the world's largest data center market, is home to more than 150 hyperscale data centers — more than 35% of all hyperscale data centers worldwide. The state is also home to hundreds of smaller data centers with many more in various planning and construction stages. These facilities bring local tax revenue and economic benefits, however, their **demand for energy has, in turn, driven up costs for utility customers and increased air pollution. At the same time, utilities have leveraged the surge in use as justification for more methane gas generation and pipeline infrastructure. Additional concerns include data centers' high water-usage rates, noise pollution and communities being shut out of the decision-making process.**

A recent Washington Post-Schar School poll that found **support for construction of nearby data centers has plunged from 69% in 2023 to 35% in 2026.** The poll also found weakening support for continuing the tax breaks for data centers that are at the heart of a budget stalemate between the Virginia Senate and House of Delegates. **The state exempts certain equipment used in data centers from the retail sales and use tax** — an exemption that lasts until June 30, 2035, under current law. **Support for that exemption plummeted from 61% in 2023 to 26% in the latest poll.**

Recent statewide polls confirm similar trends

Other recent polls are revealing similar trends in data center sentiment. A Morning Consult poll of 801 registered Virginia voters conducted on behalf of Stand.earth in April found that **Virginians are overwhelmingly concerned about the negative impacts of data center construction: 83% are concerned about rising electricity prices, 76% are concerned about water contamination and 70% are concerned about water scarcity. Seventy-eight percent of poll respondents informed of a recent study detailing the environmental and health costs associated with data centers expressed concern about air pollution caused by data centers** — including 75% of Republicans.

Eighty percent of respondents in this poll oppose the sales and use tax exemptions for data centers.

A majority of voters (52%) said, however, that they would be more likely to support a data center in their community if it were powered entirely by renewable energy with no diesel generators or on-site fossil fuel power.

“No matter where you look or who you talk to, it’s clear that the way Big Tech operates its AI data centers is extremely unpopular,” said Nathan Taft, Senior Campaigner on Stand.earth’s Fossil-Free Computing Campaign. “If companies like Microsoft and Amazon want the social license to keep building these projects that overhaul entire landscapes, they need to significantly ramp up investment in renewable energy and do right by local communities.”

Polls show increasing concerns about the impact of data centers on electric utility rates. **The Washington Post poll found that 57% of registered voters believe data centers are having a negative impact on residential utility bills** — as opposed to only 14% who believe they are helping.

A Global Strategy Group poll conducted with Potential Energy found that **voters attribute rising energy costs to demand from data centers and utilities trying to maximize their profits.** The poll found strong support for using clean energy to meet growing demand — with 70% supporting more clean energy in Virginia and 62% of Virginians agreeing that, *due to rising demand from data centers*, we need to build more clean energy sources to keep electricity costs from skyrocketing.

The GSG poll found much antipathy toward data centers: More than 6 in 10 voters had unfavorable views of big tech, AI data centers and monopoly utility companies. Two of the most popular policies tested in the poll require data centers to pay their fair share for upgrading the electricity grid (94% support) and to use clean energy and efficiency measures to meet strong energy targets (72%).

It’s not just Virginia

Negative sentiment about data centers is common among voters nationwide, not just in Virginia, polling shows. A Blue Rose Research poll conducted for Climate Power found that **national voters are especially concerned about data centers’ energy consumption and contribution to utility costs. More than twice as many voters (48%) believe the burdens of data centers outweigh the benefits as those who believe the benefits outweigh the negatives (22%).**

Most (52%) voters polled oppose the construction of data centers near where people live. And voters were more likely to support construction of new data centers powered by clean energy (53%) than those powered by fossil fuels (31%).

Morales 5-16 (Jowi Morales is a writer and journalist covering the tech beat since 2021, “AI data centers trigger massive ‘irreversible’ 76% electricity price spike in largest US region — federal watchdog demands tech giants pay for their own power infrastructure”, May 16, 2026, yahoo,
https://finance.yahoo.com/sectors/energy/articles/ai-data-centers-trigger-massive-120235475.html?guce_referrer=aHR0cHM6Ly93d3cuZ29vZ2xlLnNvbS8&guce_referrer_sig=AQAAACa25_TpCosTXK1iF9ITNL8cX0ilqqOr2VAH8UFUAOCWqTWgQYNK6st4b7QereM9FFtMv1VuGmPu9VchyYF

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*** Brackets in original

Monitoring Analytics, the federally mandated independent watchdog that keeps an eye on the critical PJM Interconnection that distributes power around the U.S., **said in a new report [PDF] that a massive 75.5% increase in power costs in the largest region of the U.S. has been directly caused by data centers** and it also blamed the regional market operator for failing to keep up with the rising demand. The price increases have been steep; wholesale electricity prices went up from \$77.78 per MWh in the first quarter of 2025 to \$136.53 per MWh in the same period of this year. "The price impacts on customers have been very large and are not reversible," the watchdog said in the report. "The price impacts will be even larger in the near term unless the issues associated with data center load are addressed in a timely manner, prior to the next BRA (base residual auction), scheduled for June 2026." Monitoring Analytics says that PJM Interconnection is trying to rewrite the rules for the capacity market and bake in data center demands into its forecasts. The watchdog is critical of this proposal as it will raise the prices for all electricity consumers, putting an unnecessary burden on households and small businesses. There is a possible solution to this problem: make data centers and other major consumers negotiate directly with power producers instead of mixing that demand with the BRA. This auction is where capacity is sold some three years ahead of when it's needed, and including data center demand in the BRA will push prices up for everyone. In fact, the report says that without the AI infrastructure built out, the PJM Capacity Market would not have seen the high prices we're experiencing now. So, by making data centers negotiate directly with power producers, this will ensure that the expanded capacity is solely shouldered by these large consumers and help keep utility bills stable for everyone else. However, this is not within the interests of PJM Interconnection. After all, by keeping massive data center loads baked into the general capacity forecast, the power auction would result in higher prices as demand moves up, but supply stays at relatively the same level. This higher cost, in turn, would then be passed on to transmission operators and local utilities, eventually making its way into the bill of the individual consumer. The increasing backlash against data center development, particularly its impact on electricity prices, has caught the attention of the federal government. In March of this year, President Donald Trump gathered some of the country's biggest AI hyperscalers in the White House and made them promise that they would "pay their own way" when it comes to AI infrastructure costs. This is, in principle, what Monitoring Analytics is pushing for: have tech companies pay for their own power — both the electricity they consume, and the infrastructure needed for it. **Unfortunately, the "ratepayer protection pledge" is nothing but a promise**, and it cannot force institutions like PJM Interconnection to not pass on the burden of cost to the average American unless Congress passes a federal law that forces the Federal Energy Regulatory Commission (FERC) to prevent cost-shifting.

Yañez-Barnuevo 26 [Miguel Yañez-Barnuevo is a project manager for the Environmental and Energy Study Institute with a special focus on energy efficiency and renewable energy policies and financing initiatives, "Data Center Power Demands Are Contributing to Higher Energy Bills" PD Feb 24 26 DOA Aug 10 26, EESI]

<https://www.eesi.org/articles/view/data-center-power-demands-are-contributing-to-higher-energy-bills>//DLin //recut RT

As data centers expand nationwide, utilities are receiving hundreds of gigawatts in

interconnection requests, with implications for the power grid and consumers. Dozens of utilities received data center requests for at least 700 gigawatts (GW) of power connection development in 2025, which is more than the 477 GW in electricity that the United States consumed in all of 2023. Even though many of these projects will never be built, the requests are still leading to a ramp-up in energy infrastructure investments, including generation facilities, transmission lines, and transformers. Utilities make decisions about building or purchasing new power in a complicated ecosystem of independent system operators (ISOs), state energy regulators, and local regulations. Electric utilities are the first to receive load-interconnection applications (or applications to connect to the grid) for proposed data centers within their service territories. Regulated investor-owned utilities are obligated to serve interconnection requests within their service territory, as set out in the "obligation to serve" or "duty to serve" legal principle. All types of utilities—regulated and non-regulated—must balance these power requests against current generation capacity, future demand projections, and reliability concerns. State energy regulatory entities (i.e., public utility commissions) approve requests from regulated utilities to increase energy rates to cover upfront investments in building new power plants, transmission lines, and related infrastructure, such as transformers. Non-regulated utilities can adjust their rates without seeking approval. An Affordability Crisis Is Brewing **To cover the investments needed to accommodate the surge in interconnection requests, many utilities are passing on the costs to their consumers through higher monthly utility bills. Utilities requested more than**

\$29 billion in rate increases in the first half of 2025, double the amount requested in the first half of 2024.

Electric rate increases were expected to affect 40 million customers nationwide last year. This is in addition to the rate increases granted by state energy commissions over the previous couple of years. Before 2019, electricity prices had been flat at around 13 cents per kilowatt-hour (kWh) for more than a decade—in part thanks to energy efficiency policies that reduced energy consumption as the overall domestic economy grew. By the end of 2025, however, average electricity prices in the United States increased to 19 cents per kWh, about 27% more than in 2019.

In states with a high concentration of data centers like Virginia, electricity prices have increased by up to 267% over the last five years. Such spikes are due to utilities needing to quickly deploy infrastructure, such as power lines and

transformers, and pay extra for market-rate energy.

Electricity prices are not expected to decrease anytime soon. According to the U.S. Energy Information Administration, residential electricity prices rose by 11.5% in 2025, outpacing inflation. Looking further ahead, prices are expected to increase by up to 40% by 2030 compared to 2025.

ERCOT 26 --- (no author, ERCOT, 6-2-2026, [Texas's electricity council] "Jun 02, 2026 Board of Directors Meeting", No Publication, [https://ercot.new.swagit.com/videos/389805?](https://ercot.new.swagit.com/videos/389805?do=9-11-2026), do=9-11-2026)

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SO WHAT IS GROWING? LET'S GET BACK TO IT.

UM, SO WHAT'S GROWING IS HERE, THE GREEN LINE AND THE BLUE LINE.

SO WHAT'S THE GREEN LINE? GREEN LINE IS NON BUILDING.

SO IF YOU LOOK, THAT'S CONSTRUCTION ACTIVITY, POWER PLANTS, UH, WATER INFRASTRUCTURE, THOSE TYPES OF, UH, YOU KNOW, ROADS AND THINGS.

AND THAT'S REALLY, AND, AND THAT, THAT GREEN LINE IS VERY VOLATILE AND IT REALLY MOVED UP SURGED LAST YEAR.

AND SO WE THINK IT'S RELATED TO THE POWER INFRASTRUCTURE CERTAINLY AND ALSO THE IMPROVEMENTS WE'RE DOING AT WATER INFRASTRUCTURE IN THE STATE AND OTHER INFRASTRUCTURE IMPROVEMENTS.

SO THAT'S A POSITIVE DEVELOPMENT, BUT IT SPIKES AND THEN IT'LL COME BACK DOWN.

UM, THE NON-RESIDENTIAL IS THE, IS THE BLUE LINE, AND THAT'S REALLY MORE WHAT I'M GONNA TALK ABOUT THAT'S REALLY BEEN TRENDING UP, UM, SINCE 20 21, 20 22.

AND THAT'S WHERE THE DATA CENTERS ARE .

SO LET'S LOOK MORE CLOSELY AT THAT DATA CENTER ACTION.

SO THAT'S THE LIGHT BLUE LINE HERE.

YOU SEE, IT'S REALLY BEEN, UM, EXPONENTIAL GROWTH IN THE CONSTRUCTION ACTIVITY AROUND DATA CENTERS.

UH, REALLY OVERALL, LAST YEAR IT WAS ABOUT 11 BILLION OR ABOUT 5.7% OF TOTAL CONSTRUCTION ACTIVITY WAS JUST DATA CENTERS.

BUT IF YOU, BY THE END OF THE YEAR, GIVEN IT WAS, WAS RISING ALL LAST YEAR, IT WAS CLOSER TO TO 9% OF TOTAL CONSTRUCTION ACTIVITY.

SO YOU THINK JUST DATA CENTERS, THAT'S A TREMENDOUS SHARE OF CONSTRUCTION ACTIVITY GROWTH, THAT'S JUST DATA CENTERS.

UM, TEXAS RATES PRETTY HIGH ON DATA CENTER CONSTRUCTION.

WE ARE NUMBER TWO IN THE NATION.

WE'RE CONSISTENTLY NUMBER TWO AFTER VIRGINIA.

UM, AND SO YOU CAN UNDERSTAND WHY IF YOU WANNA BUILD A DATA CENTER, UH, AND YOU HAVE YOUR PICK, UH, OF ANY STATE, UH, IN THE COUNTRY, THAT IT MAKES A LOT OF SENSE TO COME TO TEXAS.

CHEAP LAND PRICES, UH, YOU KNOW, EASY, RELATIVELY EASY PERMITTING, UH, AND RELATIVELY LOW ELECTRICITY PRICES THANKS TO ERCOT.

UH, SO THESE ARE ALL, UM, REASONS TO COME TO TEXAS AND YOU KNOW, FRANKLY, YOU KNOW, I LOOK AT THESE DATA AND I THINK IT COULD EVEN LIKE CATAPULT US INTO THE TOP STATE FOR DATA CENTER CONSTRUCTION.

AND I DON'T KNOW IF Y'ALL THINK THAT'S GOOD NEWS OR BAD NEWS.

CERTAINLY MORE CHALLENGES FOR OUR CO UM, SO ONE CONCERN WITH DATA CENTERS OF COURSE IS WHAT IT'S GONNA DO TO ELECTRICITY PRICES.

AND, UM, SO, SO SO FAR PRETTY GOOD.

LET ME SHOW YOU SOME.

THIS IS RETAIL ELECTRICITY PRICES.

SO IT'S NOT JUST WHOLESALE ELECTRICITY, IT'S ACTUALLY INCLUDES TRANSMISSION AND DISTRIBUTION.

SO, UM, THIS IS REALLY HOW ELECTRICITY PRICES, INFLATION ADJUSTED, HAVE DEVELOP, YOU KNOW, HAVE EVOLVED SINCE 2015.

CALIFORNIA, THAT'S NOT A DATA CENTER THING, I DON'T THINK IT'S MOSTLY TRANSMISSION AND DISTRIBUTION.

THEY HAVE HAD A LOT OF FIRES THAT HAVE DESTROYED A LOT OF THEIR INFRASTRUCTURE AROUND ELECTRICITY TRANSMISSION.

AND SO THEY'VE HAD TO REBUILD THAT.

THEY'VE FACED A LOT OF COSTS, UH, FIXED COSTS IN THEIR ELECTRICITY PRICES.

SO THAT'S REALLY, UH, BEEN DRIVING THEIR PRICES UP QUITE A BIT OVER THIS PERIOD OF TIME.

TEXAS IS THE BOTTOM LINE.

IT'S THE BLUE LINE AGAIN, AND THERE IF YOU ADJUST FOR INFLATION, YOU CAN SEE ELECTRICITY.

RETAIL ELECTRICITY PRICES ARE ACTUALLY DOWN, UH, FROM 2015.

SO WE'RE REALLY NOT SEEING THE DATA CENTER CONSTRUCTION, WHICH HAS BEEN MASSIVE PLAY OUT IN ELECTRICITY PRICES HERE YET.

UM, BUT YOU CAN SEE IN VIRGINIA A. LITTLE BIT, IT LOOKS LIKE DEFINITELY IN VIRGINIA STARTED LAST YEAR IN PARTICULAR, THAT ORANGE LINE TO MOVE UP.

WE THINK THAT'S THE NUMBER ONE STATE FOR DATA CENTER CONSTRUCTION.

WE'RE THINKING THAT'S SPILLING OVER INTO ELECTRICITY PRICES AND, AND THAT, THAT WOULD MAKE SENSE.

WE ACTUALLY HAVE SOME REALLY INTERESTING RESEARCH AT THE DALLAS FED, AND I WANTED TO SHARE THE CONCLUSIONS OF THAT RESEARCH, UH, BY MY COLLEAGUES, OWEN K AND RE TAYLOR AND THEIR CO-AUTHOR.

AND THEY'RE ACTUALLY MEASURING A LOT OF THESE THINGS EXACTLY, UH, WITH GREAT DATA THAT THEY WERE ABLE TO GET ON DATA CENTER CONSTRUCTIONS, UH, AND OPENINGS AND UH, AND ELECTRICITY PRICES.

AND THAT CERTAINLY MAKE A LOT OF ASSUMPTIONS ABOUT LOW GROWTH AND SO FORTH.

BUT, UH, BUT THEY LOOK AT WHOLE THE IMPACT OF DATA CENTERS ON WHOLESALE ELECTRICITY PRICES, AND HERE'S WHAT THEY FIND.

SO LOOKING BACKWARD, THEY FIND THAT, UH, **NEW DATA CENTERS OR DATA CENTERS IN TOTAL HAVE ADDED ABOUT 3%, UH, TO WHOLESALE ELECTRICITY PRICES IN OUR REGION IN THE ERCOT REGION, UH, IN THE PERIOD 2021 TO 2025.**

SO 3% IS, IS NOT MUCH.

UM, AND SO, AND THAT'S SIGNIFICANTLY LESS THAN THE NATIONAL AVERAGE.

SO, SO LIKE I SAID ON THE PRIOR SLIDE, DOESN'T SEEM LIKE WE'RE SEEING A LOT OF PRESSURE ON PRICES HERE YET.

FORWARD-LOOKING.

HOWEVER, **CONTINUED DATA CENTER CONSTRUCTION**

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AND OPENINGS ARE EXPECTED TO ADD ABOUT 10% TO WHOLESALE ELECTRICITY PRICES IN THE ERCOT REGION IN 2026 TO 2028.

This increase is literally life or death.

Graff 25 --- (Phd1; Ther W. Aung Michelle Graff [Michelle Graff, PhD1, She received a master's degree in environmental resource policy from George Washington University. She also holds a bachelor's degree in economics, 10/27/2025, "Energy Insecurity and Mental Health Symptoms in US Adults", No Publication,

<https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2840540>, doa9-11-2026) //MR 🐼

Results

The study sample included 1 139 607 respondents representing a weighted population size of 187 356 336 US adults (45 551 454 [24.3%] aged 18-34 years; 96 305 955 [51.4%] female, 91 050 381 [48.6%] male). Weighted measures are shown in Table 1. In terms of race and ethnicity, 9 456 479 respondents (5.0%) were Asian, 19 085 944 (10.2%) were Black, 30 125 654 (16.1%) were Hispanic, 120 709 420 (64.4%) were White, and 7 978 838 (4.3%) were of other race or ethnicity. Less than one-quarter of respondents (43 294 903 [23.1%]) had an annual income of less than \$35 000.

Table 1. Weighted Sociodemographic Characteristics of the Study Population View LargeDownload(opens in new tab) Characteristic Weighted No. (%) (N = 187 356 336)a Outcome variablesDepression symptomsb 34 082 583 (18.2) Anxiety symptomsc 43 779 826 (23.4) Energy-related SDOHInability to pay energy bill 42 029 443 (22.4) Keeping home at unsafe temperature 41 126 954 (22.0) Forgoing expenses to pay energy bill 63 009 461 (33.6) Composite energy insecurity indicatord 81 176 522 (43.3) Other SDOHUnemployed during past week 69 587 511 (37.1) Housing instability 14 242 333 (7.6) Food insecurity during past week 20 327 361 (10.8) Age, y 18-34 45 551 454 (24.3) 35-49 50 446 614 (26.9) 50-64 47 979 322 (25.6) ≥65 43 378 946 (23.2) Sex Female 96 305 955 (51.4) Male 91 050 381 (48.6) Race and ethnicity Asian 9 456 479 (5.0) Black 19 085 944 (10.2) Hispanic 30 125 654 (16.1) White 120 709 420 (64.4) Other 7 978 838 (4.3) Educational level High school or less 63 801 905 (34.1) Some college or associate's degree 55 244 376 (29.5) Bachelor's or graduate degree 68 310 055 (36.5) Annual household income, \$ <35 000 43 294 903 (23.1) 35 000-74 999 54 684 819 (29.2) 75 000-149 999 55 414 463 (29.6) >150 000 33 962 151 (18.1) Marital status Married 106 697 729 (56.9) Widowed, divorced, or separated 34 814 108 (18.6) Never married 45 844 499 (24.5) Household size, No. of persons 1-2 82 117 738 (43.8) 3-4 70 453 707 (37.6) ≥5 34 784 891 (18.6) No. of children in home None 120 991 383 (64.6) 1-2 52 136 096 (27.8) ≥3 14 228 858 (7.6) Home type Detached 1-family 127 109 668 (67.8) Attached 1-family 13 935 829 (7.4)

Apartment 35 468 175 (18.9) Otherf 10 842 664 (5.8) Home ownership Owner 131 071 314 (70.0) Renter 56 285 022 (30.0) No. of severe disabilities None 160 967 724 (85.9) 1 19 788 916 (10.6) ≥2 6 599 696 (3.5) Region Northeast 31 568 870 (16.8) South 71 642 032 (38.2) Midwest 39 210 375 (20.9) West 44 935 058 (24.0) Policy protection from utility disconnection No 158 665 456 (84.7) Yes 28 690 880 (15.3) 28 690 880 (15.3)

One-fifth of the population reported they were unable to pay an energy bill (42 029 443 [22.4%]). Similarly, **41 126 954 respondents (22.0%) kept home temperatures at unsafe or unhealthy levels due to cost concerns.** A higher prevalence of households—**63 009 461 (33.6%)—reported forgoing expenses, such as food or medicine, to pay an energy bill.** The composite energy insecurity indicator revealed **81 176 522 US adults (43.3%) reported experiencing at least 1 form of energy insecurity in the past year.** In comparison, 69 587 511 adults (37.1%) reported being unemployed during the past week and 20 327 361 (10.8%) reported being food insecure in the past week, and 14 242 333 (7.6%) noted housing instability.

Bivariate unadjusted analyses (Table 2) illustrate that experiencing any energy insecurity indicator, including the composite measure, was associated with depression and anxiety symptoms. Similarly, experiencing other SDOHs, including housing instability and food insecurity, was associated with mental health symptoms.

The same thing has happened in the past. When utility bills rose in Europe by just TWELVE CENTS a unit,

Broughel 23 James Broughel [Dr. James Broughel was a Senior Fellow at the Competitive Enterprise Institute. Dr. Broughel is an accomplished economist whose expertise lies in regulatory institutions and the impact of regulations on economic growth. He is author of the book “Regulation and Economic Growth: Applying Economic Theory to Public Policy” and is a regular columnist at Forbes, where he writes on topics such as energy and cryptocurrency], “The lethal impact of rising energy prices”, 6/26/23, CEI <https://cei.org/blog/the-lethal-impact-of-rising-energy-prices/>, DOA: 09-09-2026 //SC //recut RT

Many of us are all too familiar with the grimace we make when we open our monthly utility bills. The soaring cost of energy is a burden we all bear, often grudgingly slashing our energy use or stretching our budget thin to cover the exorbitant costs. But have you ever stopped to consider that high energy prices may not just be an annoyance, but a literal matter of life and death? The evidence paints a chilling picture.

Economists have delved into the cold realities of energy costs, investigating the link between price spikes and mortality rates. One such study was conducted by the Economist Magazine last month, focusing on European mortality rates last winter and comparing them to the figures from 2015-2019. According to the study, there were 149,000 excess European deaths last winter. At the same time, prices of electricity were up 69% and gas was up 145% compared to two years prior. No doubt Russia’s war in Ukraine, combined with tight restrictions on European energy supply, contributed to the price spike.

The implications are startling: **a rise of merely €0.10 per kwh was associated with an increase in a country’s weekly mortality rate of approximately 2.2%. Shockingly, 68,000 deaths were linked to higher energy prices.** To put that into perspective, the study attributed 59,700 excess deaths to the ongoing Covid-19 pandemic during the same time period.

The Economist suggests expensive energy discourages people from adequately heating their homes, thereby increasing the risk of cardiac and respiratory problems due to prolonged exposure to cold conditions. This should not surprise us, given a regular reported finding in statistical research is that “warmer is healthier,” meaning higher temperatures are associated with lower death rates.

The Economist study helps to demonstrate why the impact of high energy prices is cause for concern. But we also need to look at the cost of regulation, which impacts household budgets indirectly. Regulations can contribute to an increased death toll by imposing costs that eat into disposable income. As spending power dwindles, so too does the potential for spending on risk management and health-related expenses. This argument, known as the “wealthier is healthier” hypothesis, complements the idea that warmer is healthier. In fact, spending on energy is potentially a critical channel that can explain the frequently observed relationship between financial health and physical health, including mortality.

Two of my own research papers include estimates of the level of cost sufficient to produce one expected death in society. Depending on the study method, my coauthors and I found that for about every \$40 to \$115 million in costs imposed on American society, we can predict one death will occur by virtue of individuals being made poorer.

When we take a step back and look at the regulatory system as a whole, the death toll from the regulatory state is not trivial. My CEI colleague Wayne Crews has estimated that the total cost of federal regulations was just under \$2.0 trillion in 2022. Other studies put the cost even higher, in the vicinity of \$4 trillion, or even as high as \$40 trillion, annually. If we take just the two low-end estimates of cost, at \$80 million per expected death there are roughly 25,000 to 50,000 deaths annually that can be attributed to federal regulations (and this doesn’t count state and local regulations).

Those numbers may sound staggering, but not when compared to the estimated 68,000 deaths in Europe from high electricity prices during a single winter, or in the context of the approximately 3 million deaths that occur every year in the United States. This suggests regulations may explain about 0.8 to 1.7 percent of annual deaths. Furthermore, regulations probably save some lives too, which means the *net* death toll is somewhat lower than the gross toll.

It is high time that we understand and address the serious ramifications of high energy prices and regulatory costs, as they may have lethal consequences. With the mounting evidence, it becomes clear that the price we pay for energy might just be higher than what shows up on our utility bills.

States alone cannot solve.

States can regulate more strictly when local conditions demand it on TOP of the federal floor. But right now, because there is no federal floor, prices are increasing.

From **Stoel Rives** LLP, [Stoel Rives is a leading US law firm providing sophisticated business clients high quality legal services. With ten offices in Alaska, California, Idaho, Oregon, Washington, Minnesota, Utah, and Washington, D.C., Stoel Rives is a nationally recognized leader in project finance and the energy and natural resources industries. From deals and disputes to compliance and counseling, clients turn to Stoel Rives for their most complex business challenges.], no author, May 13th 20**26**, “Energy Regulatory Updates”, <https://www.stoel.com/insights/reports/energy-regulatory-updates/may-13-2026> DOA: 09/18/26 //RT

Proposed Decisions and Resolutions

Resolution E-5467. This Resolution approves San Diego Gas & Electric’s request for approval of a utility-owned energy storage contract procured to address 2026 and 2027 summer reliability. The contract is a Membership Interest Purchase Agreement with RWE AG subsidiary, RWE Clean Energy Asset Holdings, Inc., for the purchase of Westside Canal 2A, LLC, a 119-megawatt project and an associated 10-year Long-Term Services Agreement. This Resolution approves the requested relief in Advice Letter (AL) 4736-E.

Voting Meeting

The CPUC will hold a voting meeting in San Francisco, California, on Thursday, May 14, 2026, at 11 a.m. PT. The energy-related items on the agenda are below:

Item 5. Application 25-06-026 (In the Matter of the Application of Crimson California Pipeline L.P. for Authority to Increase Rates Application No. for Its Crude Oil Pipeline Services).

This decision approves the unopposed request by Crimson California Pipeline L.P. (Crimson) to increase the rates it charges for transportation of crude petroleum products on its Southern California Pipeline System by 10% beginning August 1, 2025. The Commission further authorizes the retroactive charge and collection of the difference between rates billed and the approved rate beginning August 1, 2025, including interest calculated at the 90-day commercial paper rate. This decision also denies Crimson's request in its Miscellaneous Filing for an additional 3.16% rate increase beginning August 1, 2025.

Item 8. Application 22-05-022 (Application of Pacific Gas and Electric Company for Review of the Disadvantaged Communities – Green Tariff, Community Solar Green Tariff and Green Tariff Shared Renewables Programs).

This decision adopts a process to implement the customer community renewable energy tariff adopted by Decision (D.) 24-05-065. Following the United States Environmental Protection Agency's August 2025 termination of California's Solar for All award and the resulting current lack of available external funding, this decision supersedes D.24-05-065 regarding a nonparticipating customer-funded adder. The decision also addresses Green Tariff program oversight and cost recovery, Disadvantaged Communities-Green Tariff funding and program evaluation, and program reporting for California Shared Renewables Portfolio programs.

Item 9. Application 26-01-007 (Application of Southern California Edison Company for Woolsey Fire Recovery Bond Financing Order Pursuant to Public Utilities Code Section 850).

This Financing Order grants the request by Southern California Edison Company (SCE) for authority under Public Utilities Code Sections 850 and 850.1 to issue Recovery Bonds. This Financing Order authorizes SCE to issue Recovery Bonds to recover approximately \$1.951 billion of Wildfire Expense Memorandum Account costs related to the Woolsey Fire. According to the decision, the outcomes leads to ratepayer savings of approximately \$811 million compared to traditional utility financing mechanisms on a net present value basis, with the precise amount of savings depending on several factors not known at this time, such as the terms, the number and size of the tranches, and interest rates on the Recovery Bonds.

Item 17. Rulemaking 18-07-003 (Order Instituting Rulemaking to Continue Implementation and Administration, and Consider Further Development, of California Renewables Portfolio Standard Program).

This decision denies the March 6, 2025 petition to modify D.20-08-043, filed by the Bioenergy Association of California, on procedural grounds. The petition sought to extend or remove the end date of the Bioenergy Market Adjusting Tariff and proposed other programmatic changes. According to the decision, the petition fails to meet the requirements of Rule 16.4 of the California Public Utilities Commission's Rules of Practice and Procedure because these modifications could have and should have been sought within one year of the issuance of D.20-08-043.

Item 20. Application 24-03-018 (Application of Pacific Gas and Electric Company to Recover in Customer Rates the Costs to Support Extended Operation of Diablo Canyon Power Plant from September 1, 2023, through December 31, 2025, and for Approval of Planned Expenditure of 2025 Volumetric Performance Fees).

This decision grants, in part, the petition for modification of D.24-12-033, dated October 27, 2025, filed by the Alliance for Nuclear Responsibility. Accordingly, D.24-12-033 is modified, so that the updated final Resource Adequacy (RA) Market Price Benchmark (MPB) as opposed to forecast RA MPB is used to calculate RA substitution capacity costs when assessing whether the actual costs and expenses necessary to operate Diablo Canyon Power Plant fall within the 115% of the forecasted costs under Public Utilities Code Section 712.8h(1). This approach will apply to all annual Diablo Canyon Power Plant cost forecast proceedings through 2030, not just the 2025 revenue requirement approved in D.24-12-033.

Item 24. Application 24-08-002 (Application of PacifiCorp for Approval of its 2025 Energy Cost Adjustment Clause and Greenhouse Gas-Related Forecast and Reconciliation of Costs and Revenue).

This decision approves the November 10, 2025, Joint Motion for Approval of Written Settlement filed by PacifiCorp d/b/a Pacific Power the California Farm Bureau Federation, and the Public Advocates Office at the CPUC, requesting Commission approval and adoption of the all-party partial settlement (Settlement) regarding the remaining scoped issues in PacifiCorp's 2025 Energy Cost Adjustment Clause. The Settlement resolves all the issues raised with respect to the application.

Item 25. Application 24-09-001 (Application of Southwest Gas Corporation for Authority to Increase Rates and Charges for Gas Service in California, Effective January 1, 2026).

This decision grants the uncontested Joint Motion to Adopt a Settlement Agreement, filed by Southwest Gas Corporation (Southwest Gas), the Public Advocates Office, and Small Business Utility Advocates. The Settlement Agreement resolves certain issues material to Southwest Gas' application for authorization to increase rates and charges for gas service in California, effective January 1, 2026.

Item 26. Application 24-07-020 (Application of Pacific Gas and Electric Company (PG&E) for Approval of Peak Day Supply Standard Pursuant to Decision 23-11-069).

This decision approves a proposed settlement (Settlement Agreement) reached between PG&E and the solitary protestant, The Utility Reform Network (TURN), regarding PG&E's use of a Peak Day Gas Supply Standard (PDSS) for the winter seasons in years 2024/2025 through 2026/2027. PG&E and TURN jointly seek Commission approval of their proposed Settlement Agreement, which resolves all issues in this proceeding. The decision approves the use of the proposed PDSS as set forth in the proposed Settlement Agreement and the parties Joint Motion for Approval of a Settlement Agreement for the period requested.

Upcoming Events
Rule 17 Backbilling Workshop.

The Energy Division will host a virtual workshop on May 18, 2026, at 10 a.m. to facilitate the development of an Investor-Owned Utilities and Community Choice Aggregators consensus-based proposal on how to address uncollected revenue from the application of Rule 17 backbilling limitations for both bundled and unbundled customers. An agenda and any additional materials will be distributed closer to the workshop.

Workshop for Developing a Distribution System Operator (DSO)-led DER Orchestration Framework.

On Thursday, May 21, 2026, from 10 a.m.-5 p.m., the CPUC will host a Workshop in Track 2 of the Distributed Energy Resources (DER) proceeding regarding developing a DSO-Led DER Orchestration framework. The workshop will be held both in-person at the CPUC Auditorium and remote. Additional details are available on the High DER proceeding docket, R.21-06-017.

CALIFORNIA ENERGY COMMISSION (CEC)
2026 Integrated Energy Policy Report (IEPR) Update

The CEC has set the following preliminary schedule for the 2026 IEPR. The schedule is available on the 2026 IEPR Update page.

Task / Event	Date
• Final 2026 IEPR Update Scoping Order released	April 2026
• Adopt order instituting informational proceeding	May 2026
• Public workshops on specific topics	May–December 2026
• Release draft 2026 IEPR Update	October 2026
• Release proposed 2026 IEPR Update	January 2027
• Adopt 2026 IEPR Forecast	January 2027
• Adopt 2026 IEPR Update	February 2027

The CEC's 2026 IEPR Update workshop schedule (which remains subject to change) is set forth below. According to the workshop schedule, all workshops will be held via Zoom and will run from May through December 2026, as follows:

May 20, 2026: Staff Workshop on Forms & Instructions, 2 p.m.-4 p.m. PT. Written comments are due to the Docket Unit by 5 p.m. PT on June 3, 2026, and should reference Docket No. 26-IEPR-02.

June 19, 2026: Commissioner Workshop on Geothermal Energy (Part 1), 10 a.m.-5 p.m. PT

June 30, 2026: Commissioner Workshop on Energy Equity & Environmental Justice, 1 p.m.-5 p.m. PT

July 9, 2026: Commissioner Workshop on Geothermal Energy (Part 2), 10 a.m.-5 p.m. PT

August 20, 2026: Commissioner Workshop on Forecast Inputs and Assumptions, 1 p.m.-5 p.m. PT

August 31, 2026: Commissioner Workshop on Load Modifier Inputs and Assumptions, 9 a.m.-4:30 p.m. PT

November 12, 2026: Commissioner Workshop on Load Modifier Draft Results, 10 a.m.-5 p.m. PT

December 14, 2026: Commissioner Workshop on Overall Forecast Results, 10 a.m.-5 p.m. PT

Draft 2025 IEPR

On April 23, 2026, the CEC issued a Notice of Availability and Request for Comments regarding the Draft 2025 Integrated Energy Policy Report. As set forth in the July 2025 Scoping Order and as summarized in the Notice of Availability, the Draft 2025 IEPR contains in-depth discussions regarding advancing clean energy deployment; electricity and gas forecast; and provides a load-shift goal update. Written comments are due to Docket No. 25-IEPR-01 before 5 p.m. on May 15, 2026.

Resource Planning and Reliability

On May 4, 2026, the CEC hosted a workshop regarding summer energy reliability. Staff from the CEC, California ISO, CPUC, CALFIRE, Western Electricity Coordinating Council, California Department of Water Resources, among others, provided information regarding anticipated summer conditions. Written comments are due to the Docket Unit by 5 p.m. PT on May 18, 2026.

Workshop on SB 237 Assessment

The CEC announced that it will host a workshop on May 18, 2026, to evaluate the recommendations and strategies put forward by Vice Chair Siva Gunda in the June 27, 2025 letter to California Governor Newsom regarding Senate Bill 237. A link to the Draft SB 237 Assessment, titled "Supporting the Transition Away from Petroleum Fuels," is available [here](#). The workshop will be remote-access only and login details are available [here](#), along with an agenda.

Electric Program Investment Charge (EPIC)

On April 2, 2026, the CEC circulated a "Save the Date" for the CEC's annual EPIC Symposium indicating that the event is scheduled for Tuesday, September 29, 2026, at the California Natural Resources Agency building (715 P Street) in Sacramento, California. More details will be provided when made available by the CEC.

CEC Business Meetings

The next CEC business meeting is scheduled for May 13, 2026, from 10 a.m.-3 p.m. PT. The agenda and supporting materials are available [here](#).

CALIFORNIA AIR RESOURCES BOARD (CARB)

Concepts for Potential Carbon Regulations

CARB posted a preliminary compilation of potential and partial regulation concepts to establish the new Carbon Capture, Removals, Utilization, and Storage (CCUS) Program through a document titled Concepts for Potential Regulations for Establishing the Carbon Capture, Removals, Utilization, and Storage Program. Under Senate Bill 905, CARB seeks public input as part of its pre-rulemaking activity for the development of the CCUS Program through an upcoming rulemaking. CARB released draft rulemaking concepts that identify potential definitions for CCUS and Carbon Dioxide Removal (CDR), as well as initial regulatory concepts for reporting, monitoring, financial responsibility, and protocols that further support CCUS and CDR inclusion in California's climate policies. CARB will accept written comments through June 5, 2026, at 11:59 PM PST.

MINNESOTA PUBLIC UTILITIES COMMISSION (MPUC)

At its May 15, 2026, weekly agenda meeting, the MPUC will consider Xcel Energy's petition for approval of its Large General Time of Day Service Customer and Large Peak Controlled Time of Day Service Tariffs, which will involve consideration of many elements, including approval of associated rider revisions, the use of an Incremental Cost Test, a determined demand size threshold, modifications to the minimum ESA term length, and the establishment of a separate customer class for super-large customers (Docket No. 25-289). The MPUC hearing will represent the first consideration of a state utility's large customer tariff following passage of Minn. Stat. Section 216B.1622, which directs the MPUC to define what constitutes a "very large customer" and set the characteristics of the customer class by December 15, 2026.

PACIFIC NORTHWEST (OPUC, WUTC, BPA)

Bonneville Power Administration (BPA)

On May 11, 2026, BPA held a workshop to review its Network Integration Transmission Service business practice manual (BPM), the clean version of which is [here](#). More information about changes to its BPM is [here](#). Additionally, on May 12, 2026, BPA's Transmission Services held a system planning meeting to provide customers and interested parties with the opportunity to discuss and share feedback on the 2027 planning studies and development of the 2027 plans of service. More information is [here](#).

Oregon Public Utilities Commission (OPUC)

On May 7, 2026, the OPUC approved Portland General Electric's (PGE) data center rate framework, pursuant to House Bill 3546, which requires the OPUC to establish a separate class for data centers, cryptocurrency miners, and similar large load customers. The OPUC's Order No. 26-154 approved several provisions of PGE's proposed data center class, such as minimum transmission and generation demand requirements, a fixed distribution charge for customers with at least 1 MW of load, contract terms, and exit fees. The order also established a dedicated interconnection queue for data center customers and focused on shifting growth costs to large load users, while also adding safeguards for emissions compliance. A number of resolutions adopted within the order were done by stipulation, such as the agreement to open an investigation to address new load connection costs. Local coverage is [here](#).

Washington Utilities and Transportation Commission (WUTC)

On May 14, 2026, the WUTC will hold its regular public meeting to discuss Avista Corporation's petition for an order approving the 2027-2030 demand and resource supply forecast as directed by the Climate Commitment Act (UE-260067), and Northwest Natural Gas Company d/b/a NW Natural's tariff revisions to Tariff WN U-6, Schedule 308 (Washington Climate Commitment Act Recovery Mechanism), which would decrease revenue by \$9.2M or 8.4% (UG-260133). The full agenda is [here](#).

FLORIDA

Florida Large Load Cost Responsibility

Florida Governor Ron DeSantis signed legislation that requires large load customers (those with a monthly peak load of 50 MW or more) to bear their own full cost of service to avoid shifting those costs onto the general body of ratepayers. The bill notes that such cost includes connection, incremental transmission, incremental generation, and other infrastructure costs; operations and maintenance expenses; and any other costs required to serve a large load customer.

The bill states that the Florida Public Service Commission may approve public utility tariffs that include industry-accepted ratemaking and other financial tools, including:

Contributions in aid of construction or other required customer infrastructure investments that may be returned, in whole or in part, to such customers over time;

Demand charges;

Incremental generation charges;

Financial guarantees;

Minimum load factors;

Take-or-pay provisions or provisions requiring payment for contracted capacity, regardless of a large load customers' actual electricity use or demand; and

Minimum period of service contract requirements, including early termination fees or other fees for violation of such contracts.

Federal Agencies

FEDERAL ENERGY REGULATORY COMMISSION (FERC)

FERC to Act on Large Load Interconnection Docket

FERC issued a statement that it will take action by June 2026 on the Advance Notice of Proposed Rulemaking (ANOPR) proceeding initiated by the U.S. Secretary of Energy. The ANOPR requested that FERC consider changes to large load interconnection, including increasing demand from data centers.

Independent System Operators (ISO) and Regional Transmission Organizations (RTO)

CALIFORNIA INDEPENDENT SYSTEM OPERATOR (CAISO)

Stakeholder Initiatives: Upcoming Meetings and Deadlines

Save the Date: Resource Interconnection Standards Fair

The CAISO will host the annual Resource Interconnection Standards Fair on Tuesday August 11, 2026. The fair will take place to provide stakeholders with the opportunity to learn more about the CAISO's interconnection process and recent changes affecting Cluster 16, in preparation of the Cluster 16 window opening on October 1, 2026. Meeting registration details can be found [here](#).

Extended Day-Ahead Market Congestion Revenue Allocation Phase 2

The CAISO revised the schedule for the upcoming working group meetings for Phase 2 of the Extended Day-Ahead Market Congestion Revenue Allocation initiative. The working group meetings will take place May 11, June 2, and June 24, 2026. More details about the meetings can be found [here](#).

PJM INTERCONNECTION LLC (PJM)
Maryland Complaint Against PJM Cost Allocation Rules

The Maryland Office of People's Counsel filed a complaint against PJM asking FERC to find that PJM's existing cost allocation rules for baseline transmission projects are unjust and unreasonable because they impose costs on Maryland's electric customers not commensurate with benefits received. The Office of People's Counsel argues that over the next decade **Maryland customers will pay about \$1.6 billion [more] in additional [electric] costs for transmission projects, driven primarily by the energy needs of data centers located outside of Maryland** resulting from PJM's hybrid load-ratio share and solutions-based distribution factor transmission cost allocation methodology for assigning transmission cost responsibility across sub-regions of PJM. The Office of People's Counsel asks for a refund effective date 60 days from filing.

ISO NEW ENGLAND (ISO-NE)
Energy Forecast Report

ISO-NE published its annual Forecast Report of Capacity, Energy, Loads, and Transmission (CELT Report), which for the first time included behind-the-meter battery energy storage systems (BTM BESS). ISO-NE stated that although these resources are not visible to ISO-NE in its control room, the ISO keeps track of them to monitor demand. ISO-NE notes that while there is currently about 111 MW of BTM BESS, less than 1 MW in size across the ISO, ISO-NE expects about 173 MW of BTM BESS to be added over the next 10 years, mostly in Massachusetts.

[1] Per the CPUC's Rules of Practice and Procedure Rule 14.3, comments on proposed decisions are due 20 days after issuance of the proposed decision, and reply comments are due five days thereafter. Comments on draft resolutions are due 20 days after the draft resolution appears in the CPUC's daily calendar, per Rule 14.5.

A federal floor would prevent this.

Our second contention is eliminating emissions.

Data centers release huge amounts of carbon dioxide.

By Alex de **Vries-Gaob**, [Alex de Vries-Gao is a PhD candidate at the VU Amsterdam Institute for Environmental Studies and the founder of Digiconomist, a research company dedicated to exposing the unintended consequences of digital trends. His research focuses on the environmental impact of emerging technologies and has played a major role in global discussion regarding the sustainability of blockchain technology and artificial intelligence.], January 9th, 20**26**, "The carbon and water footprints of data centers and what this could mean for artificial intelligence", Science Direct, <https://www.sciencedirect.com/science/article/pii/S2666389925002788> DOA: 08/16/26 //RT

Discussion and conclusion

Although there are ways to estimate the global power demand of AI systems, it remains challenging to quantify the associated carbon and water footprints. The lack of distinction between AI and non-AI workloads in the environmental reports of data center operators means it is only possible to assess the environmental impact of AI workloads by approximating them through data centers' general performance metrics. The environmental disclosure of tech companies is, however, often insufficient to assess even the total data center performance of these companies. In such cases, company-wide metrics may be the best available information, but these values may not accurately represent the data center performance, depending on the (undisclosed) weight of the data centers within the respective companies' operations. Furthermore, company-wide metrics are not always available, and some companies do not disclose any environmental impact data. Specifically, embedded water consumption in purchased electricity is rarely disclosed at any level.

The shortcomings in the environmental disclosure of data center operators could be remedied with new policies mandating the disclosure of additional metrics. Because the environmental impact of data centers is growing rapidly, the urgency for transparency in the tech sector is also

79.7 million tons of CO2 emissions in 2025, while the water footprint could reach 312.5–764.6 billion L.

There is strong variation in the carbon and water intensity of local power grids; thus, estimation accuracy would greatly improve if companies disclosed the specific locations where AI systems are operated and the magnitudes of these operations. Additionally, PUE and WUE values should be disclosed for each of these locations.

Declaration of interests

From **I**nternational **E**nergy **A**gency [The International Energy Agency (IEA) works with governments and industry to shape a secure and sustainable energy future for all.], no author found, April 10th 20**25**, “Energy and AI”, <https://www.iea.org/reports/energy-and-ai/executive-summary>, DOA: 08/16/26 //RT

[illegible]

The widespread adoption of existing AI applications could lead to emissions reductions that are far larger than emissions from data centres – but also far smaller than what is needed to address climate change. We estimate that emissions reductions from the broad application of existing AI-led solutions to be equivalent to around 5% of energy-related emissions in 2035. Various barriers to AI adoption will need to be overcome to unlock these gains. Rebound effects – for example from modal shifts away from public transport to autonomous cars – could undercut some of these benefits. AI can be a tool in reducing emissions, but it is not a silver bullet and does not remove the need for proactive policy.

With energy and tech now on a journey together, collaboration is key

The tech sector and energy industry are more intertwined than ever before. There are large uncertainties on the path ahead, but these should not get in the way of concerted action. Delivering the energy for AI, and seizing the benefits of AI for energy, will require even deeper dialogue and collaboration between the tech sector and the energy industry. Along the way, there will be risks to manage. The IEA will continue to provide data and robust analysis to inform decision making and help the energy and technology sectors be better prepared as the adoption of AI unfolds.

To combat this, data centers need renewables.

Gnibga 23 [Wedan Emmanuel Gnibga received the engineering degree in embedded electronic systems and systems control from the National School of Applied Sciences in Marrakech, Morocco, in 2020. He is currently working toward the PhD degree in computer science with the University of Rennes 1. His research interests focus on energy efficiency, energy management optimization, Cloud Computing and distributed systems. "Renewable Energy in Data Centers: The Dilemma of Electrical Grid Dependency and Autonomy Costs" 8/23/2023, American edge project

<https://americanedgeproject.org/five-reasons-why-american-needs-to-win-the-race-for-artificial-intelligence/>] DOA: 9/18/2026 //SA //recut RT

Integrating larger shares of renewables in data centers' electrical mix is mandatory to reduce their carbon footprint.

However, as they are intermittent and fluctuating, renewable energies alone cannot provide a 24/7 supply and should be combined with a secondary source. Finding the optimal infrastructure configuration for both renewable production and financial costs remains difficult. In this article, we examine three scenarios with on-site renewable energy sources combined respectively with the electrical grid, batteries alone and batteries with hydrogen storage systems. The objectives are first, to size optimally the electric infrastructure using combinations of standard microgrids approaches, second to quantify the level of grid utilization when data centers consume/ export electricity from/to the grid, to determine the level of effort required from the grid operator, and finally to analyze the cost of 100% autonomy provided by the battery-based configurations and to discuss their economical viability. Our results show that in the grid-dependent mode, 63.1% of the generated electricity has to be injected into the grid and retrieved later. In the autonomous configurations, the cheapest one including hydrogen storage leads to a unit cost significantly more expensive than the electricity supplied from a national power system in many countries.

However, right now, data centers rely on fossil fuels.

Gorey 25 [Jon Gorey is a staff writer at the Lincoln Institute of Land Policy, where he connects readers with the Institute's research and explores how land policy can help solve some of the world's most pressing social, economic, and environmental problems. An award-winning freelancer, he has written extensively about housing, travel, and other subjects, most frequently for the Boston Globe. He's also held communications roles at major media outlets and nonprofits, including MSN Money and WGBH, and is the author of Home Buying 101, published by Simon & Schuster. He's received more than a dozen awards from the National Association of Real Estate Editors (NAREE), including the President's Gold Award for Best Freelance Collection in 2019. He's a graduate of the Newhouse School of Public Communications at Syracuse University. "Data Drain: The Land and Water Impacts of the AI Boom" 10/17/2025

<https://www.lincolnst.edu/publications/land-lines-magazine/articles/land-water-impacts-data-centers/> DOA: 09/18/2026]//SA

Locating data centers in cooler, wetter regions can help reduce the amount of water they use and the impact of their freshwater withdrawals. And yet roughly two-thirds of the data centers built since 2022 have been located in water-stressed regions, according to a Bloomberg News analysis, including hot, dry climates like Arizona.

The warm water-cooling system at a Sandia Labs data center in Albuquerque, New Mexico. The data center earned LEED Gold certification for efficiency in 2020. Credit: Bret Latter/Sandia Labs via Flickr CC.

It's not just cooling the server rooms and chips that consumes water. About **half of the electricity currently used by US data centers comes from fossil fuel power plants**, which themselves use a lot of water, as they heat up steam to turn their massive turbines.

And the millions of microchips processing all that information? By the time they reach a data center, each chip has already consumed thousands of gallons of water. Manufacturing these tiny, powerful computing components requires “ultrapure” treated water to rinse off silicon residue without damaging the chips. It takes about 1.5 gallons of tap water to produce a gallon of ultrapure water, and the typical chip factory uses about 10 million gallons of ultrapure water each day, according to the World Economic Forum—as much as 33,000 US households.

As communities consider the benefits and risks of data center development, consumers might consider our own role in the growth of data centers, and whether our use of AI is worth the price of the water, power, and land it devours.

There could be important uses for artificial intelligence—if it can be harnessed to solve complex problems, for instance, or to improve the efficiency of water systems and electric grids.

There are clearly superfluous uses, too. A YouTube channel with 35 million subscribers, for example, features AI-generated music videos ... of AI-generated songs. The MIT Technology Review estimates that, unlike simple text queries, using AI to create video content is extremely resource-heavy: Making a five-second AI-generated video uses about as much electricity as running a microwave nonstop for over an hour.

In order to protect the environment, data centers must decrease these carbon emissions.

From U.S. **E**nvironmental **P**rotections **A**gency (**.gov**), [The mission of EPA is to protect human health and the environment.], no author found, no author quals, May 17th, 20**23**, “U.S. EPA’s ENERGY STAR Program Develops Energy-Saving Guidance for Co-Location Data Centers in Collaboration with Equinix and Iron Mountain”,

<https://www.epa.gov/newsreleases/us-epas-energy-star-program-develops-energy-saving-guidance-co-location-data-centers> DOA: 08/16/26 //RT

WASHINGTON - Today, the U.S. Environmental Protection Agency (EPA) announced the release of a new best practices guide to help customers of shared data centers, also known as co-locations, save energy and reduce their carbon footprint. Developed with input from global data centers, Equinix and Iron Mountain, EPA created this guidance to address common issues like energy waste and unnecessary operational costs.

“Data centers are central to meeting our ever-growing need for data processing and storage,” said EPA Administrator Michael S. Regan.

“Prioritizing energy efficiency in these data centers will allow our economy to expand to meet these needs while reducing the greenhouse gas emissions associated with them.”

Critical to society and the digital economy, co-location data centers are large data centers that rent out space within their facilities to third parties to store their servers and network equipment. This service is popular among businesses that may not have the capacity or resources to manage their own private data center but still require secure facilities for storing, processing, and protecting their data. Moreover, co-location data centers provide services that are more efficient than businesses operating their own data center.

In 2018, data centers were estimated to consume about 1% of all global energy – and despite the success of data center efficiency over the past decade, the pace of digitalization and energy consumption has been increasing, and more businesses are needing to rent space to store their data. Many quick and easy efficiency improvements – such as replacing old, inefficient hardware – have been achieved, and data center managers are now looking to improve setup and infrastructure, which accounts for roughly 38% of data center energy consumption, to optimize efficiency. This new guidance on best practices was developed to make those improvements by addressing such issues as:

Airflow management, which can reduce energy use by 15% when setup is optimized for adequate air flow.

Optimal operating temperature, which can reduce cooling costs by 4% for every degree that the air inlet temperature is raised.

Proper cabling, which can improve airflow, reduce stress on IT equipment and ensure that debris and inhibited airflow does not reduce the lifespan of hardware.

Separation of hot and cold aisles, which can reduce energy consumption by 10-35% depending on setup.

Server efficiency, which has continued to improve with the recent release of a new ENERGY STAR server specification. The latest requirements expanded the program scope to include storage-heavy servers and include criteria levels three times more stringent than what was included previously. The new ENERGY STAR specification was developed in close collaboration with industry allies to further push the bar on efficiency for server products – with the result that servers are expected to perform 38% more efficiently than standard models in order to earn the label. You can find this guidance on best practices for co-location data centers on the ENERGY STAR website, and find more guidance on data center efficiency at ENERGY STAR's Data Centers page.

About ENERGY STAR

ENERGY STAR® is the government-backed symbol for energy efficiency, providing simple, credible, and unbiased information that consumers and businesses rely on to make well-informed decisions. Thousands of industrial, commercial, utility, state, and local organizations rely on their partnership with EPA to deliver cost-saving energy efficiency solutions. Since 1992, ENERGY STAR and its partners helped American families and businesses avoid more than \$500 billion in energy costs and achieve more than 4 billion metric tons of greenhouse gas reductions. More background information about ENERGY STAR's impacts.

That's key, since climate change kills people, and every fraction of a degree of warming that can be limited saves lives.

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Abstract

When attempting to quantify future harms caused by carbon emissions and to set appropriate energy policies, it has been argued that the most important metric is the number of human deaths caused by climate change. Several studies have attempted to overcome the uncertainties associated with such forecasting. In this article, approaches to estimating future human death tolls from climate change relevant at any scale or location are compared and synthesized, and implications for energy policy are considered. Several studies are consistent with the "1000-ton rule," according to which a future person is killed every time 1000 tons of fossil

carbon are burned (order-of-magnitude estimate). **If warming reaches or exceeds 2 °C this century, mainly richer humans will be responsible for killing roughly 1 billion mainly poorer humans through anthropogenic global warming**, which is comparable with involuntary or negligent

manslaughter. On this basis, relatively aggressive energy policies are summarized that would enable immediate and substantive decreases in carbon emissions. The limitations to such calculations are outlined and future work is recommended to accelerate the decarbonization of the global economy while minimizing the number of sacrificed human lives.

Keywords: carbon emissions; greenhouse gas emissions; global catastrophic risk; climate change; energy policy; human mortality; climate genocide

1. Introduction

Despite repeated and ever more serious warnings [1] from the scientific community, global greenhouse gas (GHG) emissions and carbon dioxide (CO₂) concentrations in the atmosphere continue to increase as fossil fuel combustion increases [2]. It is now established with 95% confidence that anthropogenic global

climate destabilization is occurring [3,4] and has potentially irreversible negative repercussions for global environment and the social and economic welfare of humanity [5,6]. Yet, despite these well-known facts, there has been a reluctance to apply aggressive energy policies to eliminate fossil fuel combustion. The reasons are both political (involving conflicting interests) and psychological (involving abstractions that may not be directly perceptible).

Climate change causes human deaths in diverse ways, which can be divided into direct, intermediate, indirect, and their interactions. Direct mortal effects of climate change include heat waves, which have already caused thousands of human deaths [7,8,9] by a combination of heat and humidity (wet-bulb temperature > 35 °C, such that the human body is physically unable to cool itself with perspiration). Intermediate causes of death (between direct and indirect) involve crop failures, droughts, flooding, extreme weather, wildfires, and rising seas. Crop failures [10,11] can exacerbate existing socially-constructed global hunger and starvation resulting in tens of thousands of human deaths [12,13,14]. More frequent or severe droughts [15,16,17] can lead to more frequent or severe wildfires [18,19,20] that also cause human deaths. Droughts lead to contaminated water, diseases, and deaths from dehydration [21,22]. The 2022 IPCC Report (6th Assessment Report) predicted that drought would displace 700 million people in Africa by 2030 [23]. Climate change can cause flooding and consequent destruction of property and crops, which also drives hunger and disease [24]. Climate change drives sea level rise and the resultant submersion of low-lying coastal areas and shoreline erosion [25,26], saltwater intrusion [26,27], storm damage to coastlines, and exacerbated flood risks [18,28,29,30]. These dangers are life-threatening for billions of people in coastal cities who face the prospect of forced migration [31]. Climate change increases extreme weather events (e.g., hurricanes), which indirectly kill and cause damage worth billions of dollars [32,33,34]. Extreme weather events kill in many ways including, for example, electric grid failures and power outages [35,36] that reduce access to electric-powered medical care. Indirectly, climate change increases the probability of conflict [37,38,39,40,41]. As the number of climate refugees increases [42,43,44], countries further from the equator might increasingly refuse to offer asylum. In a worst-case scenario, social collapse is possible [45,46]. For humanity, the most important consequence of climate change will be human deaths. The value of human life may be considered the foundation of all human values, as reflected by the universal legal importance of crimes such as murder, manslaughter, and genocide [47]. Therefore, when attempting to quantify the future harms caused by carbon emissions, the number of human deaths caused by climate change may be the most important metric. Philosopher John Nolt explained [48]:

Climate change will cause large numbers of casualties, perhaps extending over thousands of years. Casualties have a clear moral significance that economic and other technical measures of harm tend to mask. They are, moreover, universally understood, whereas other measures of harm are not. Therefore, the harms of

climate change should regularly be expressed in terms of casualties by such agencies such as IPCC's Working Group III, **in addition** to whatever other measures are used. Casualty estimates should, furthermore, be used to derive estimates of casualties per emission source up to a given date. Such estimates would have wide margins of error, but they would add substantially to humanity's grasp of the moral costs of particular greenhouse gas emissions. "Casualties" can be deaths, injuries, or illness. In the medical literature these complexities are normally dealt with using the disability-adjusted life year (DALY), which is a measure of overall disease burden, expressed as the number of years lost due to ill-health, disability or early death [49,50]. DALYs have been found to be a useful tool for quantifying human damage from environmental pollution [51]. Since the World Bank first published the DALY concept in 1993, the use of life expectancy estimates, disability weights, age weighting, and discounting has evolved [52]. Unfortunately, as researchers in many fields have adjusted the equations used for estimating the DALY and have increasingly challenged assumptions underlying these calculations, comparing DALY estimates across different studies has become difficult. More concerning are studies that argue that the conceptual and technical basis for DALYs is flawed, its assumptions and value judgments are questionable as well as implications of age-weighting and discounting [53,54,55]. The stakes are high for the assumptions that underlie a DALY because it is often used for cost-effectiveness analysis, which is used for funding allocations [56]. Rather than attempt to resolve that debate, and in agreement with Nolt [48], this review will focus only on human deaths caused by climate change, because they are more easily defined and categorically more serious than injuries or illnesses. In future scenarios of interest, death tolls are also likely to correlate with rates of injury and illness, making each of the three values an approximate measure or predictor of the other. This review aims to provide a clear guide for energy policy makers, which have historically not fully considered the ramifications of AGW on human deaths. In the following text, published approaches to quantifying climate change in terms of human death tolls will first be summarized. After that, and on that basis, recommendations will be made for energy policy which are intended to be approximately applicable at every scale and in every geographic location.

2. Approaches to Quantifying Carbon Emissions with Human Deaths

2.1. The 1000-ton Rule

The 1000-ton rule says that a future person is killed every time humanity burns 1000 tons of fossil carbon. It is derived from a simple calculation: burning a trillion tons of fossil carbon will cause 2 °C of anthropogenic global warming (AGW) [57,58], which in turn will cause roughly a billion future premature deaths spread over a period of very roughly one century [59]. On the assumption that 2 °C of warming is either already inevitable (given the enormous political and economic difficulties of achieving a lower limit) or intended (given that the business plans of big fossil fuel industries make it inevitable), it can be concluded that burning 1000 tons of fossil carbon causes one future premature death.

The numbers "one billion" (for the total death toll at 2 °C) and "one thousand" (for the amount of carbon that needs to be burned to cause one death) are both very approximate (both are hardly more than order-of-magnitude estimates), but also consistent with diverse evidence and arguments:

Before 2022, humans burned roughly 0.6 trillion tons of fossil carbon, causing a global temperature increase of roughly 1.2 °C. Incidentally, about the same amount of carbon is currently part of living things on this planet (550 billion T) [60].

The carbon budget for 2 °C of AGW is about one trillion tons [61]. Thus, if humanity burns that amount altogether, the global mean surface temperature will rise by 2 °C. A more exact estimate is not necessary, because predicted death tolls will inevitably be even more approximate.

About five trillion tons of fossil carbon are available in the Earth's crust. If humanity collectively burned all of that, global mean surface temperature would increase by up to 10 °C relative to the pre-industrial era [57,58] and could threaten human extinction [62]

Dividing one trillion by one billion, one thousand tons is the amount of carbon that needs to be burned today to cause a future premature death in the future: 1000 tons.

It has been clear for a decade or more [63] that the final death toll due to AGW will be much greater than 100 million, or one million per year for a century—an extreme best case if current death rates from AGW miraculously remained constant at about one million per year (a level that may have already have reached). Conversely, the final death toll in a 2 °C warming scenario will certainly be much less than 10 billion, which is the predicted global human population in 2100 in the absence of AGW [64]. Although climate change clearly represents a global catastrophic risk to food supplies [65], only a small minority are suggesting that 2 °C of warming could cause human extinction [66]. Warming of well over 2 °C, however, could indeed cause natural climate feedbacks to get out of control, leading eventually to human extinction [66]. Between these extreme boundaries, it is likely more than 300 million ("likely best case") and less than 3 billion ("likely worst case") will die as a result of AGW of 2 °C. That prediction is consistent with detailed predictions of climate science summarized by the World Health Organization and their probable consequences for human mortality [67].

2.2. Convergent Evidence for the 1000-ton Rule

Although AGW is a global concern, some studies have looked specifically at a single country's emissions (USA) to illustrate the methods used. The 1000-ton rule is roughly consistent with two such independent studies from different academic disciplines—philosophy and economics. The authors of those studies arrived at their estimates of future death tolls by different methods but came to similar conclusions. First, an American philosopher John Nolt [68] concluded that the carbon emissions of the average American are causing the death or suffering of one or two future people. That is because the average American, in the course of a lifetime, causes (by her or his personal choices or participation in regular social structures) 1840 metric tons of CO₂ equivalent to be emitted. This corresponds to about 500 tons of carbon, so according to the 1000-tonne rule, those emissions are enough to kill half of a future person.

A more recent attempt at quantifying future deaths in connection with specific amounts of carbon was published by Bressler [69]. Coining an economically oriented term “mortality cost of carbon”, he claimed that “for every 4434 metric tons of CO₂ pumped into the atmosphere beyond the 2020 rate of emissions, one person globally will die prematurely from the increased temperature” [70]. His predictions were confined to deaths from extreme heat when wet-bulb temperature exceeds skin temperature (35 °C). He acknowledged the importance of other causes of death in connection with AGW such as “infectious disease, civil and interstate war, food supply, and flooding”, but considered them difficult to quantify—although taken together they may cause more deaths than direct heat. His estimate is consistent with the 1000-tonne rule, according to which one future person dies (for one of many reasons linked to AGW) whenever roughly 3700 tons of CO₂ are emitted.

2.3. The Temperature Niche

Xu and colleagues [71] have argued that “accounting for population growth projected in the SSP3 scenario, each degree of temperature rise above the current baseline roughly corresponds to one billion humans left outside the temperature niche, absent migration”. That is consistent with the 1000-ton rule if it is assumed that long-term survival outside the ecological niche is unlikely. Specifically, for every additional degree of warming beyond 2 °C, roughly an additional billion deaths will be caused, leading to human extinction at very roughly 10 °C of warming [59]. At 2 °C of warming, roughly 2 billion people would find themselves outside the climate niche as defined by Lenton et al. [72]. If it is assumed that global warming will be the ultimate cause of death for half of those people (e.g., due to the inability to migrate), then 2 °C of warming will cause a billion deaths.

Ecologist George Evelyn Hutchinson has shown that a species faces extinction if it leaves its ecological niche [73], but that does not apply to humans with access to energy reserves. Humans without access to the capital to provide for those energy reserves, however, will die, particularly if limited resources from climate change result in ‘climate wars’ [74].

2.4. Marginal Carbon Emissions-Related Deaths

The estimates made in Section 2.1, Section 2.2 and Section 2.3 are very rough but provide a useful rule of thumb for gaging a first approximation. The 1000-ton rule makes it clear that there is a marginal human death cost to every amount of warming, no matter how small. Thus, **every 0.1 °C degree of**

warming can be expected to cause 100 million deaths. Similarly, every 0.001 °C of warming will cause

a million deaths. If humanity misses the 2 °C target or any of the more granular goals to stop ‘dangerous climate change’ [75], which appears likely according to AI models [76], rather than relax and accept it, all efforts to reduce carbon emissions can be viewed as lifesaving.

3. Results: Carbon-Related Deaths to Guide Energy Policy

3.1. Large Numbers and the Millilife

Understanding the large variability in the number of human deaths from climate change requires an accessible unit of measure for carbon footprints that is easy to understand and may be used to set energy policy to help accelerate carbon emissions reductions. Using the “1000-tonne rule” as a basis it is possible to convert any carbon footprint to human lives. A millilife is defined as a measure of intrinsic value that is equivalent to one thousandth of the value of human life. A millilife is roughly the same as a month in the richest countries, assuming a lifespan of about 80 years, but it is closer to half a month in the poorest countries, in which the lifespan is closer to 40 years.

The 1000-ton rule says that a millilife is destroyed when a ton of fossil carbon is burned. For example in Canada, which has some of the highest yearly carbon emissions per capita in the world at around 19 tons of CO₂ or 5 tons of carbon per person [77], roughly 5 millilives are sacrificed by an average person each year. As the average Canadian lives to be about 80, he/she sacrifices about 400 millilives (0.4 human lives) in the course of his/her lifetime, in exchange for a carbon-intensive lifestyle. Canada is one of the world's largest economies, currently ranked 9th [78]. Whereas the average gross domestic product globally is US\$12,183/person, Canada generates US\$51,988/person, more than 4 times the global average [78]. Canadians are thus not only wealthy by global standards but many already live outside of the human temperature niche. Climate change will likely benefit the Canadian economy in some sectors (e.g., agriculture and tourism), causing GDP to increase by up to 0.3% (\$9 billion/year) [79]. Even if some areas prosper, however, others will face major challenges that could cause great economic harm (e.g., increased forest fires [80]). By contrast, the Global South will suffer the most negative impacts of climate change [81,82,83].

Parncutt assumed [59] for the purpose of the 1000-ton rule that an average future AGW-victim in a developing country will lose half of a lifetime or 30–40 life-years, as most victims will be either very young or very old. If the average climate victim loses 35 life-years (or 13,000 life-days), a millilife corresponds to 13 days. Stated in another way: if a person is responsible for burning a ton of fossil carbon by flying to another continent and back, they effectively steal 13 days from the life of a future poor person living in the developing world. If the traveler takes 1000 such trips, they are responsible for the death of a future person.

This concept can also be applied to large-scale energy decisions. For example, the Adani Carmichael coalmine in Queensland, Australia, is currently under construction and producing coal since 2021. Despite massive protests over several years, it will be the biggest coalmine ever. Its reserves are up to 4 billion tons of coal, or 3 billion tons of carbon. If all of that was burned, the 1000-tonne rule says it would cause the premature deaths of 3 million future people. Given that the 1000-tonne rule is only an order-of-magnitude estimate, the number of caused deaths will lie between one million and 10 million. Currently, 10 million tons of coal are being mined per year [84], equivalent to 7.5 million tons of carbon or 7500 sacrificed human lives. Many of those who will die are already living as children in the Global South; burning Carmichael coal will cause their future deaths with a high probability. Should energy policy allow that to occur?

Thus, to reduce prices and protect the environment we're proud to affirm.